



香港中文大學(深圳)
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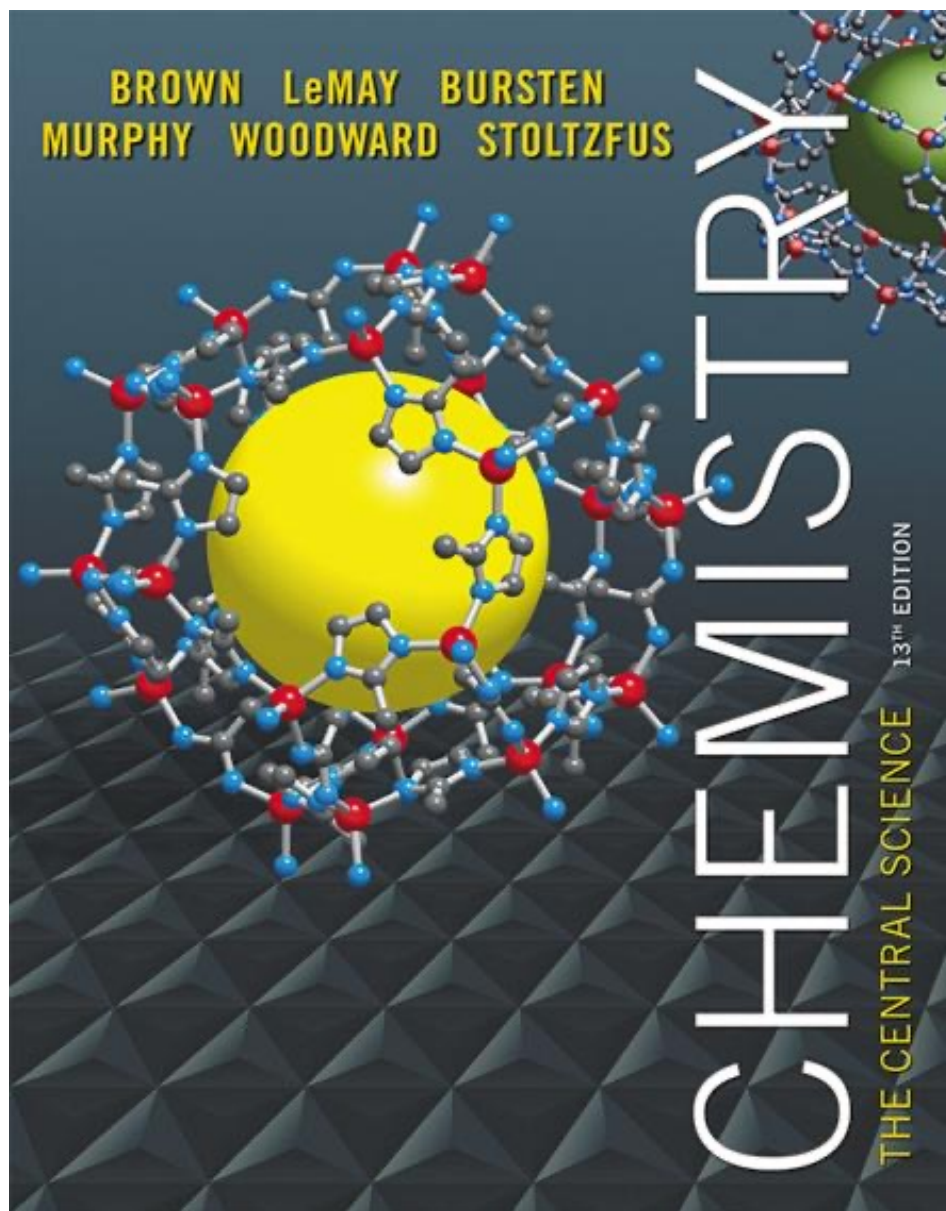
General Chemistry CHM1001

Thermodynamics (Ch. 19)

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Nov. 6th, 2025



Lecture Presentation

Chapter 19

Chemical Thermodynamics

Thermochemistry



First Law of Thermodynamics

- You will recall from Chapter 5 that energy cannot be created or destroyed.
- Therefore, the total energy of the universe is a constant.
- Energy can, however, be converted from one form to another or transferred from a system to the surroundings or vice versa.



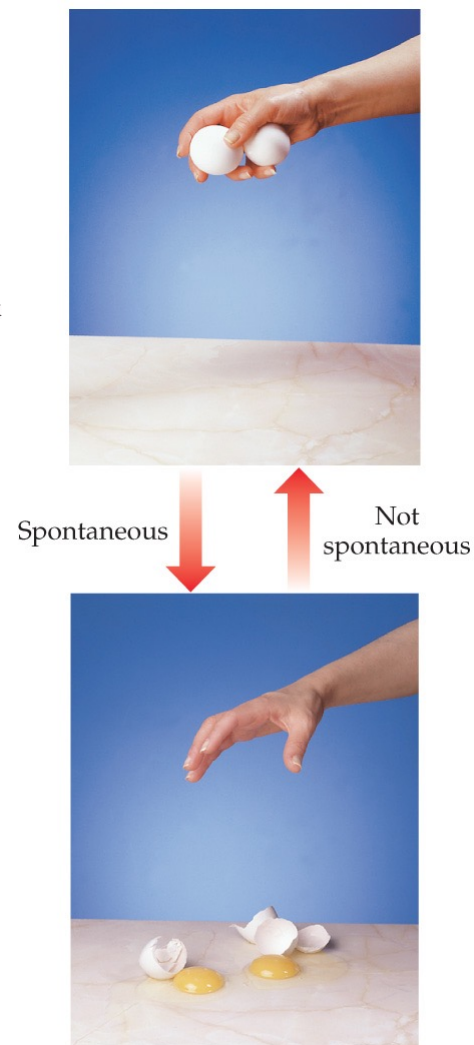
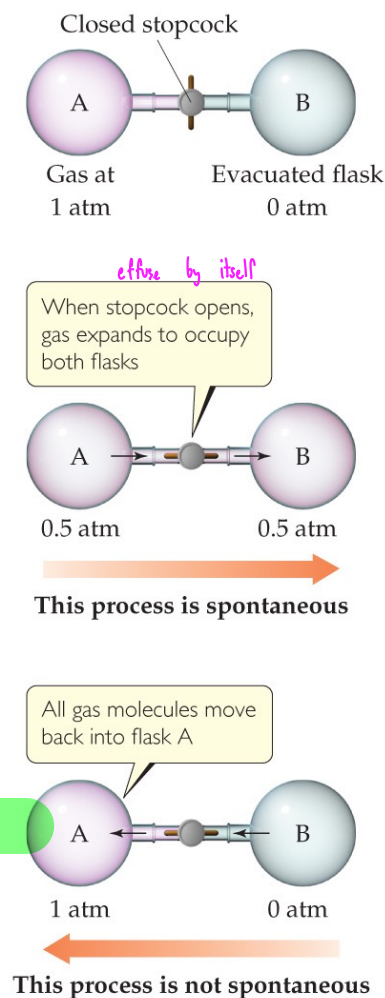
Enthalpy/Entropy

- Enthalpy is the heat absorbed by a system during a constant-pressure process.
- Entropy is a measure of the randomness in a system.
- Both play a role in determining whether a process is spontaneous.



Spontaneous Processes

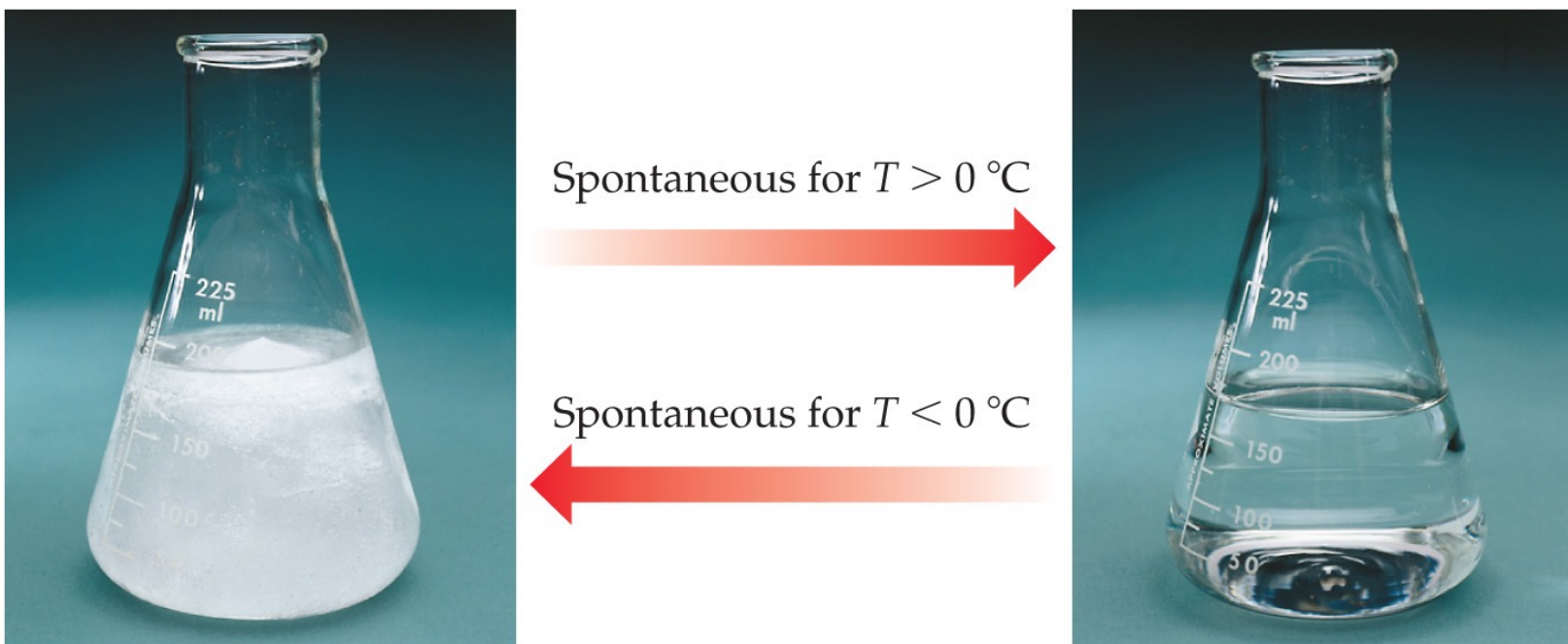
- **Spontaneous processes proceed without any outside assistance.**
- The gas in vessel A will spontaneously effuse into vessel B, but it will *not* spontaneously return to vessel A.
- Processes that are **spontaneous in one direction** are **non spontaneous in the reverse direction.**



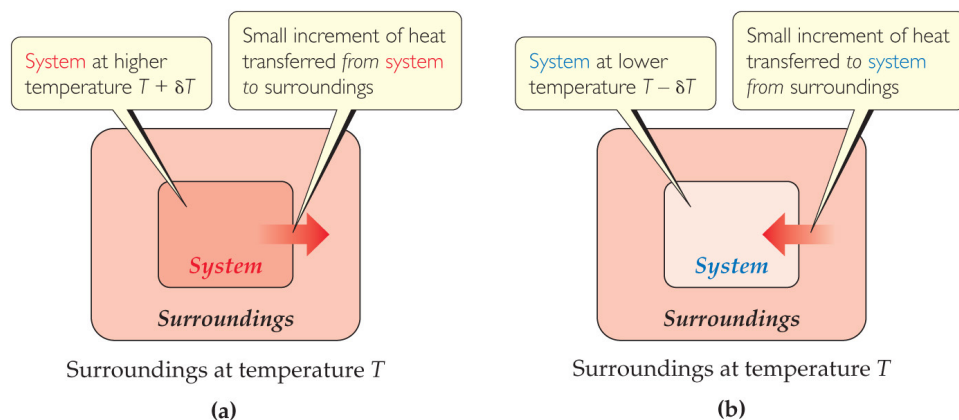
Thermochemistry

Experimental Factors Affect Spontaneous Processes

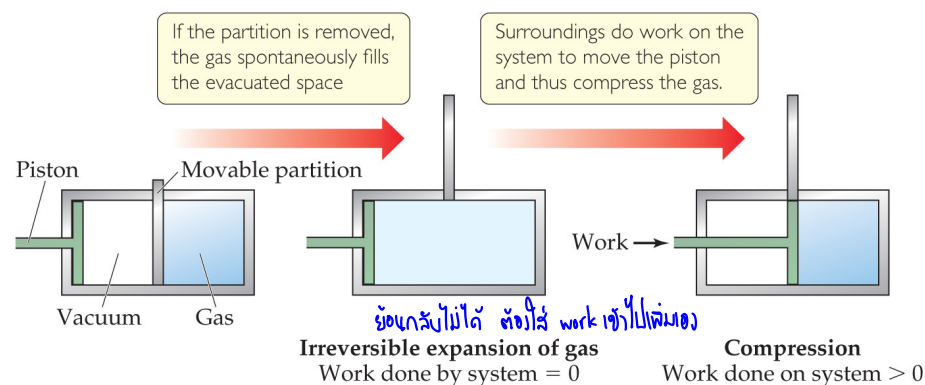
- Temperature and pressure can affect spontaneity.
- An example of how temperature affects spontaneity is ice melting or freezing.



Reversible and Irreversible Processes



Reversible process: The system changes so that the system and surroundings can be returned to the original state by exactly reversing the process. This maximizes work done by a system *on* the surroundings.



Irreversible processes cannot be undone by exactly reversing the change to the system or cannot have the process exactly followed in reverse. Also, any spontaneous process is irreversible!

Thermochemistry

Entropy

→ State function
 $q, w = \text{path func}$

- **Entropy** can be thought of as a measure of the randomness of a system.
- It is a state function:

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

- It can be found by heat transfer from surroundings at a given temperature:

$$\Delta S = \frac{q_{\text{rev}}}{T} \quad (\text{constant } T)$$



$$\Delta S = \frac{q_{\text{rev}}}{T} \quad (\text{constant } T)$$

1. Entropy as a measure of “disorder”

$$\Delta H = q \quad (\text{Const } P)$$

- Entropy represents the “degree of disorder,” “spread of energy,” or “freedom of possible states” of a system.
- Heat increases the kinetic energy of particles, making the system more random. → That’s why **heat transfer** is the fundamental cause of changes in entropy.

2. Why we divide by temperature

- If you add the same amount of heat q , in a **low-temperature system** the relative change in each particle's energy is large, so the disorder increases significantly.
- In a **high-temperature system**, particles are already moving vigorously, so the same q produces a smaller increase in disorder.
- Therefore, the increase in entropy depends not only on the amount of heat added but also on the **temperature** of the system at that moment.



Second Law of Thermodynamics

- The entropy of the universe increases in any spontaneous processes.
- This results in the following relationships:

Reversible Process:

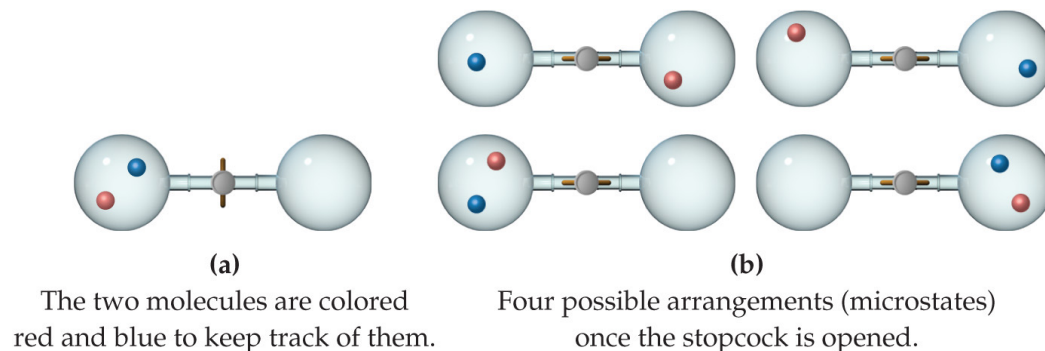
$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$$

Irreversible Process:

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$



Entropy on the Molecular Scale



- Boltzmann described entropy on the molecular level.
- Gas molecule expansion: Two molecules are in the apparatus above; both start in one side. What is the likelihood they both will end up there? $(1/2)^2$
- If one mole is used? $(1/2)^{6.02 \times 10^{23}}$! (No chance!)
- Gases spontaneously expand to fill the volume given.
- Most probable arrangement of molecules: approximately equal molecules in each side



Statistical Thermodynamics

- Thermodynamics looks at bulk properties of substances (the *big* picture).
- We have seen what happens on the molecular scale.
- How do they relate?
- We use statistics (probability) to relate them. The field is called statistical thermodynamics.
- **Microstate**: A single possible arrangement of position and kinetic energy of molecules

Thermochemistry



Boltzmann's Use of Microstates

- Because there are so many possible microstates, we can't look at every picture.
- W represents the number of microstates.
- Entropy is a measure of how many microstates are associated with a particular macroscopic state.
- The connection between the number of microstates and the entropy of the system is:

$$S = k \ln W$$



Entropy Change

- Since entropy is a state function, the final value minus the initial value will give the overall change.
- In this case, an increase in the number of microstates results in a positive entropy change (more disorder).

$$\Delta S = k \ln W_{\text{final}} - k \ln W_{\text{initial}} = k \ln \frac{W_{\text{final}}}{W_{\text{initial}}}$$



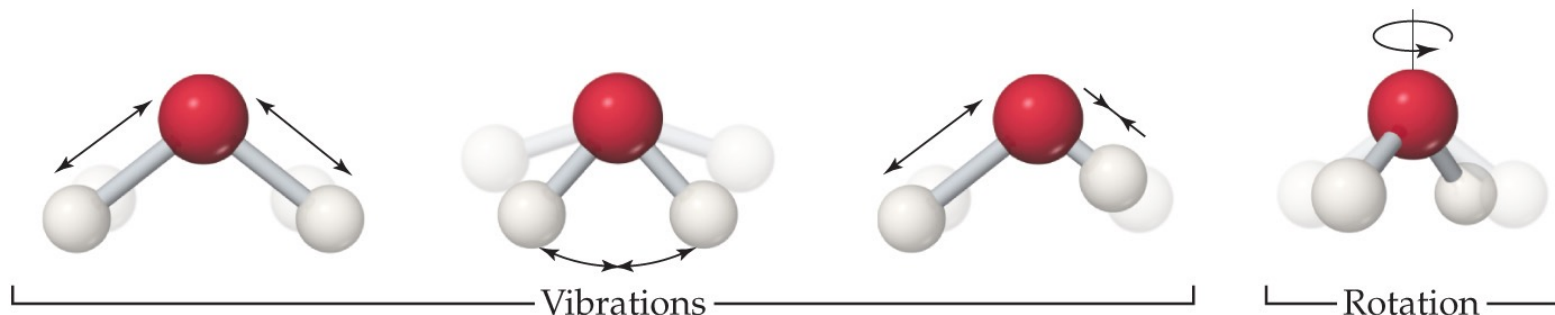
Effect of Volume and Temperature Change on the System

- If we increase volume, there are more positions possible for the molecules. This results in more microstates, so increased entropy.
- If we increase temperature, the average kinetic energy increases. This results in a greater distribution of molecular speeds. Therefore, there are more possible kinetic energy values, resulting in more microstates, increasing entropy.



Molecular Motions

- Molecules exhibit several types of motion.
 - **Translational:** Movement of the entire molecule from one place to another
 - **Vibrational:** Periodic motion of atoms within a molecule
 - **Rotational:** Rotation of the molecule about an axis
- Note: More *atoms* means more microstates (more possible molecular motions).



Thermochemistry



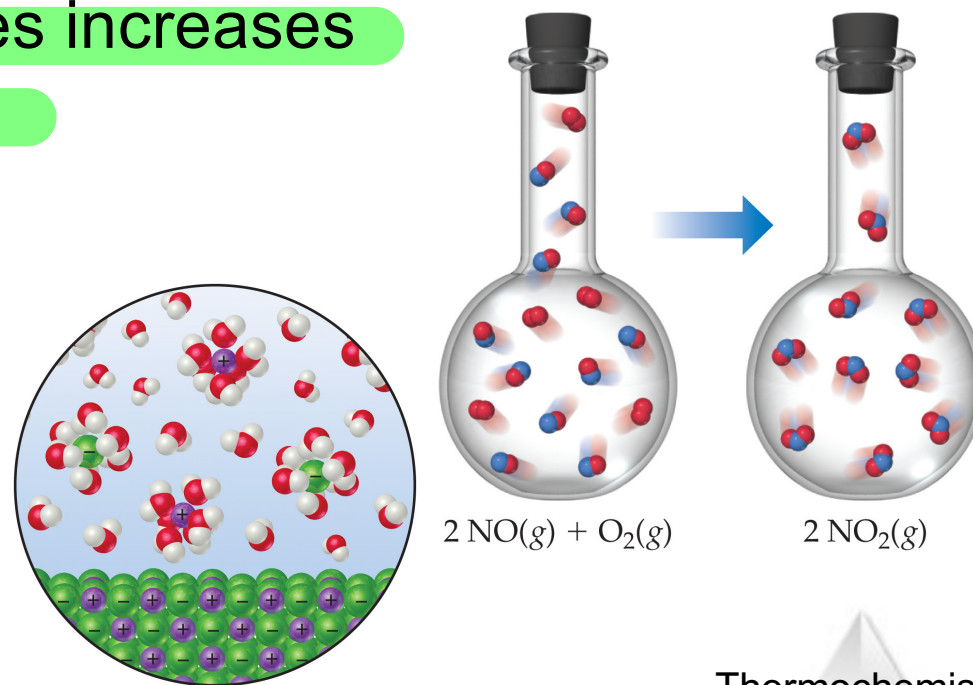
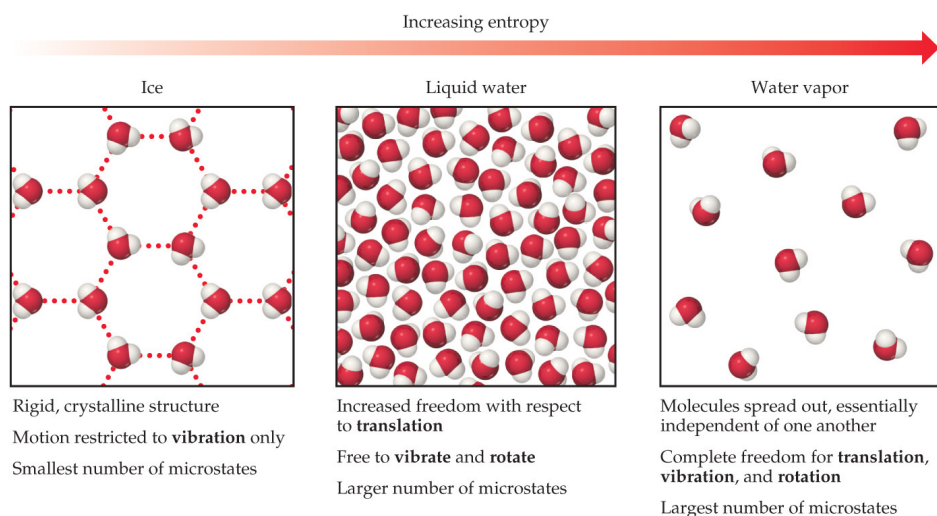
Entropy on the Molecular Scale

- The number of microstates and, therefore, the entropy tend to increase with increases in
 - temperature.
 - volume.
 - the number of independently moving molecules.



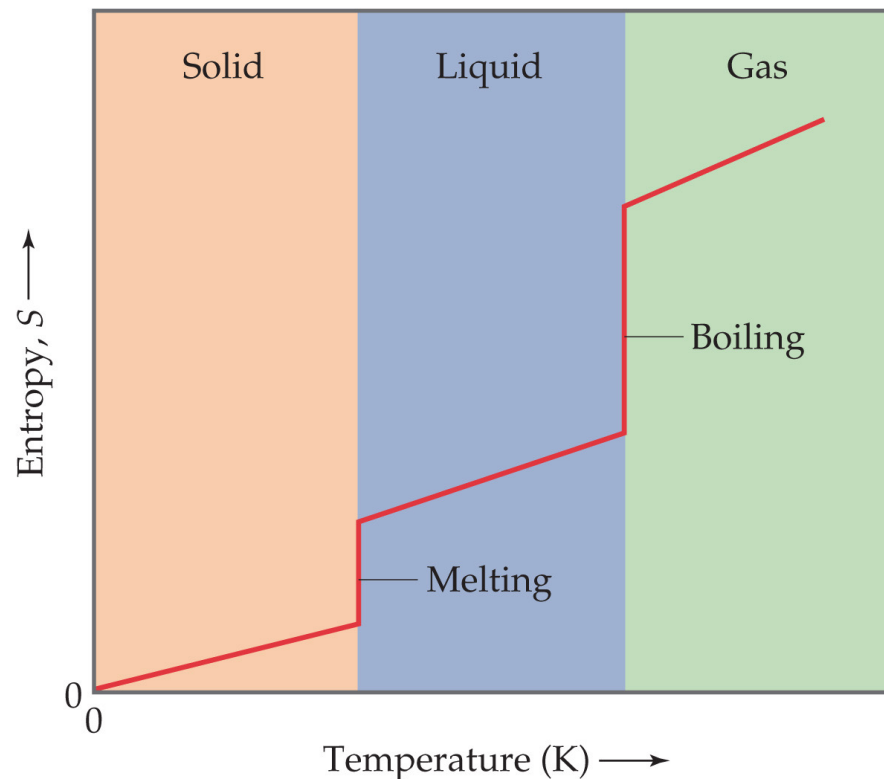
Entropy and Physical States

- Entropy increases with the freedom of motion of molecules.
- $S(g) > S(l) > S(s)$
- Entropy of a system increases for processes where
 - gases form from either solids or liquids.
 - liquids or solutions form from solids.
 - the number of gas molecules increases during a chemical reaction.



Thermochemistry

Third Law of Thermodynamics



- The entropy of a pure crystalline substance at absolute zero Kelvin is 0.
- Consider all atoms or molecules in the perfect lattice at 0 K; there will only be *one* microstate.
- $S = k \ln W = k \ln 1 = 0$

Standard Entropies

- The reference for entropy is 0 K, so the values for elements are *not* 0 J/mol K at 298 K.
- Standard molar enthalpy for gases are generally greater than liquids and solids. (Be careful of size!)
- Standard entropies increase with molar mass.
- Standard entropies increase with number of atoms in a formula.

Table 19.1 Standard Molar Entropies of Selected Substances at 298 K

Substance	S° (J/mol·K)
H ₂ (g)	130.6
N ₂ (g)	191.5
O ₂ (g)	205.0
H ₂ O(g)	188.8
NH ₃ (g)	192.5
CH ₃ OH(g)	237.6
C ₆ H ₆ (g)	269.2
H ₂ O(l)	69.9
CH ₃ OH(l)	126.8
C ₆ H ₆ (l)	172.8
Li(s)	29.1
Na(s)	51.4
K(s)	64.7
Fe(s)	27.23
FeCl ₃ (s)	142.3
NaCl(s)	72.3

Entropy Changes

Entropy changes for a reaction can be calculated in a manner analogous to that by which ΔH is calculated:

$$\Delta S^\circ = \sum n \Delta S^\circ (\text{products}) - \sum m \Delta S^\circ (\text{reactants})$$

where n and m are the coefficients in the balanced chemical equation.



Entropy Changes in Surroundings

- Heat that flows into or out of the system changes the entropy of the surroundings.
- For an isothermal process

$$\Delta S_{\text{surr}} = \frac{-q_{\text{sys}}}{T}$$

$\Delta S \uparrow$ if $q_{\text{sys} \rightarrow \text{surr}} (q_{\text{sys}} < 0)$
 $\Delta S \downarrow$ if $q_{\text{surr} \leftarrow \text{sys}} (q_{\text{sys}} > 0)$

- At constant pressure, q_{sys} is simply ΔH° for the system.

$$\Delta S_{\text{surr}} = \frac{\text{Change of enthalpy} \quad -\Delta H_{\text{sys}}}{T} \quad [\text{at constant P}]$$



Entropy Change in the Universe

- The universe is composed of the system and the surroundings.
- Therefore,

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

- For ^(Irreversible)spontaneous processes

$$\Delta S_{\text{universe}} > 0$$

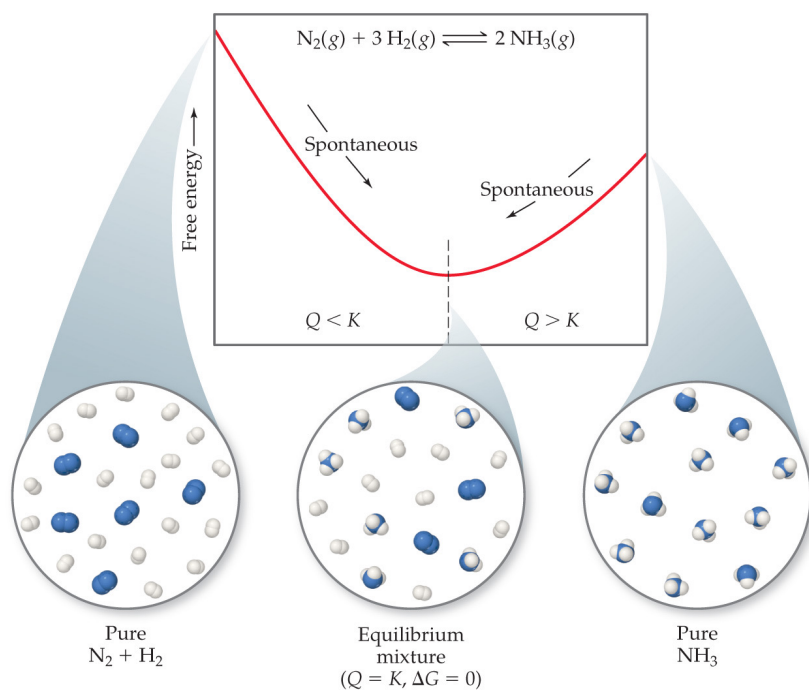


Total Entropy and Spontaneity

- $\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$
- Substitute for the entropy of the surroundings:
 - $\Delta S_{\text{universe}} = \Delta S_{\text{system}} - \Delta H_{\text{system}}/T$
- Multiply by $-T$:
 - $-T\Delta S_{\text{universe}} = -T\Delta S_{\text{system}} + \Delta H_{\text{system}}$
- Rearrange:
 - $-T\Delta S_{\text{universe}} = \Delta H_{\text{system}} - T\Delta S_{\text{system}}$
- Call $-T\Delta S_{\text{universe}}$ the **Gibbs Free Energy (ΔG)**:
 - $\Delta G = \Delta H - T\Delta S$
Focus just on system



Gibbs Free Energy



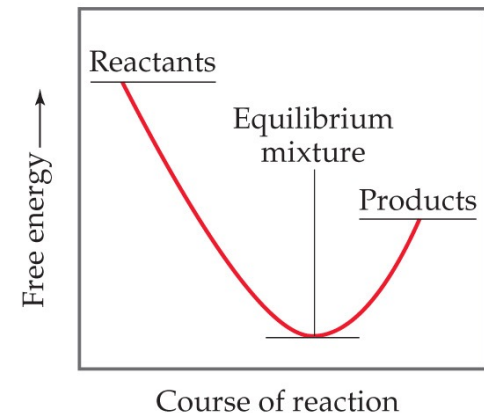
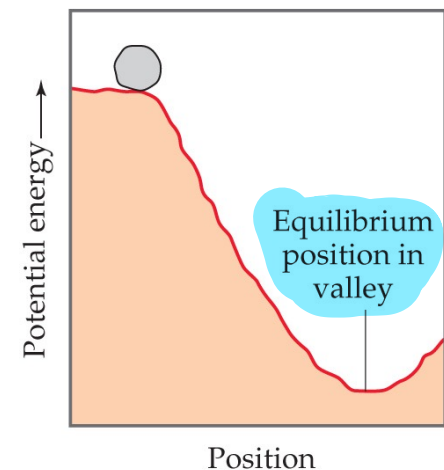
1. If ΔG is negative, the forward reaction is spontaneous.
 $\Delta S_{\text{Universe}} > 0$
2. If ΔG is 0, the system is at equilibrium.
3. If ΔG is positive, the reaction is spontaneous in the reverse direction.
 $\Delta S_{\text{Universe}} < 0$

Standard Free Energy Changes

Analogous to standard enthalpies of formation are standard free energies of formation, ΔG_f° :

$$\Delta G^\circ = \sum n\Delta G_f^\circ(\text{products}) - \sum m\Delta G_f^\circ(\text{reactants})$$

where n and m are the stoichiometric coefficients.



Thermochemistry

Free Energy Changes

Table 19.3 How Signs of ΔH and ΔS Affect Reaction Spontaneity

ΔH	ΔS	$-T\Delta S$	$\Delta G = \Delta H - T\Delta S$	Reaction Characteristics	Example
–	+	–	–	Spontaneous at all temperatures	$2 \text{O}_3(g) \longrightarrow 3 \text{O}_2(g)$
+	–	+	+	Nonspontaneous at all temperatures	$3 \text{O}_2(g) \longrightarrow 2 \text{O}_3(g)$
–	–	+	+ or –	Spontaneous at low T ; nonspontaneous at high T	$\text{H}_2\text{O}(l) \longrightarrow \text{H}_2\text{O}(s)$
+	+	–	+ or –	Spontaneous at high T ; nonspontaneous at low T	$\text{H}_2\text{O}(s) \longrightarrow \text{H}_2\text{O}(l)$

- How does ΔG change with temperature?
- $\Delta G = \Delta H - T\Delta S$
- Since reactions are spontaneous if $\Delta G < 0$, the sign of enthalpy and entropy and the magnitude of the temperature matters to spontaneity.



Free Energy and Equilibrium

Under any conditions, standard or nonstandard, the free energy change can be found this way:

$$\Delta G = \Delta G^{\circ} + RT \ln Q$$

(Under standard conditions, concentrations are 1 *M*, so $Q = 1$ and $\ln Q = 0$; the last term drops out.)

Free Energy and Equilibrium

- At equilibrium, $Q = K$, and $\Delta G = 0$.
- The equation becomes

$$0 = \Delta G^\circ + RT \ln K$$

- Rearranging, this becomes

$$\Delta G^\circ = -RT \ln K$$

or

$$K = e^{-\Delta G^\circ / RT}$$

